MILITARY SPECIFICATION

DIISOPROPYL HYDROGEN PHOSPHITE, TECHNICAL

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one grade of diisopropyl hydrogen phosphite (also known as diisopropyl phosphite) for use as an anti-wear additive in lubricants.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

STANDARDS

Military

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.

MIL-STD-129 - Marking for Shipment and Storage.

MIL-STD-290 - Packaging, Packing and Marking of Petroleum and Related Products.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

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2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM) STANDARDS

D 1193 - Reagent Water

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal Agencies.)

3. REQUIREMENTS

3.1 Physical and chemical characteristics. The diisopropyl hydrogen phosphite shall conform to the applicable physical and chemical characteristics of table I when tested as specified therein.

Table I. Physical and chemical characteristics

Characteristic	Requirement	Paragraph
Appearance	Clear and free from suspended matter	4.5.1
Color	To pass test	4.5.2
Odor	Pleasant	4.5.3
Phosphorus, percent by weight	18.4 to 18.7	4.5.4
Monoisopropyl dihydrogen phosphite, percent by weight	1.00 maximum	4.5.5
pH	2.2 to 2.5	4.5.6
Chloride, percent by weight	0.020 maximum	4.5.7
Specific gravity ($25^{\circ}/4^{\circ}C$)	0.980 to 1.002	4.5.8
Refractive index ($N_D 20^{\circ}C$)	1.406 to 1.410	4.5.9

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to specified requirements.

4.2 Classification of Inspection. The examination and testing of the diisopropyl hydrogen phosphite shall be entirely by quality conformance inspection.

4.3 Quality Conformance Inspection. The quality conformance inspection shall consist of an examination for acceptability of quality control methods used by the manufacturer, and examination of samples of filled containers (4.4.3) for conformance to packaging, packing, and marking requirements, and examinations and tests of samples (4.4.2) for all the requirements specified in Section 3.

4.4 Quality Conformance Sampling. The quality conformance sample shall consist of a sample for tests (4.4.2) and a sample for examination of filled containers (4.4.3). Samples shall be labelled completely with information identifying the purpose of the sample, name of product, specification number, lot and batch number, date of sampling, and contract number.

4.4.1 Batch or Lot. An indefinite number of containers of identical size and type, offered for acceptance at one time, and filled with a homogeneous mixture of material manufactured by a single plant run (not exceeding 24 hours) through the same processing equipment, with no change in ingredient material. In the event the process is a batch operation, each batch shall constitute a lot.

4.4.2 Samples for Tests. The sample for tests shall consist of two containers of diisopropyl hydrogen phosphite taken at random from each lot of diisopropyl hydrogen phosphite offered or processed for delivery under a contract or order. The lot shall be unacceptable if a sample fails to meet any of the test requirements specified.

4.4.3 Sample for Examination of Filled Containers. A random sample of filled containers and a sample of shipping containers fully prepared for delivery shall be selected from each lot of the blend in accordance with MIL-STD-105 at inspection level I and acceptable quality level (AQL) = 2.5 percent defective.

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4.5 Tests. Water in accordance with ASTM D 1193, Type III and reagent grade chemicals shall be used throughout the tests. Where applicable, blank determinations shall be run and corrections applied where significant. Tests shall be conducted as follows:

4.5.1 Appearance. Visually examine the specimen for clarity and suspended matter.

4.5.2 Color. Prepare a standard potassium dichromate solution by dissolving 0.0030 gm. of potassium dichromate in 1 liter of water. Compare the color of 10 ml of the specimen with 10 ml of the standard potassium dichromate solution in Nessler tubes. The color of the specimen shall be no darker than the color of the standard potassium dichromate solution.

4.5.3 Odor. Waft the vapor from the mouth of the bottle and smell carefully

4.5.4 Phosphorus.

4.5.4.1 Reagents.

4.5.4.1.1 Ammonium molybdate reagent. Prepare solution No. 1 by transferring 100 gm. of molybdic acid (85% MoO_3) to a 600-ml beaker containing 240 ml of water. Mix and add 140 ml of ammonium hydroxide while stirring vigorously. When dissolution is complete, filter through a Whatman No. 40 filter paper, add 60 ml of nitric acid, and cool. Prepare solution No. 2 by adding 400 ml of nitric acid to 960 ml of water in a 2-liter beaker and cooling. Add solution No. 1 to solution No. 2 while stirring constantly. Add 0.1 gm. of diammonium hydrogen phosphate $\{(NH_4)_2PO_4\}$ and let stand at least 24 hours. Filter through a Whatman No. 40 filter paper before using. This reagent will keep for 2 weeks.

4.5.4.1.2 Standard sodium hydroxide solution (0.3N). Standardize against potassium acid phthalate.

4.5.4.1.3 Standard nitric acid (0.15N). The equivalence of this solution in terms of standard sodium hydroxide solution is determined in the method.

4.5.4.1.4 Phenolphthalein indicator solution (1%). Dissolve 1 gm. of phenolphthalein in 100 ml of ethyl alcohol (95%).

4.5.4.2 Procedure. Pipet approximately 1 ml of the diisopropyl hydrogen phosphite (equivalent to about 1 gm. of sample) into a tared 10-ml beaker. Weigh the beaker plus sample immediately and calculate the exact weight of the sample by difference. Wash the sample into a 250-ml beaker with 1 or 2 ml of water from a wash bottle and then rinse the 10-ml beaker with a total of 25 ml of nitric acid (added in 3 portions). Add 1 ml of bromine and allow to stand 2 hours or more. Add 10 ml of perchloric acid (72%) and warm gently at the edge of the hot plate for 30 minutes. Evaporate to fumes of perchloric

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acid at moderate heat, and fume for about 5 minutes. Cool and add about 50 ml of water, cool again and dilute to 250 ml in a volumetric flask. Pipet a 15-ml aliquot (equivalent to about 0.06 gm. of sample) into a 300-ml Erlenmeyer flask and add 35 ml of water, 4 ml of perchloric acid, and 5 ml of nitric acid. Add 150 ml of ammonium molybdate reagent at one stroke and swirl immediately. Insert a tight-fitting rubber stopper and immediately shake for 30 seconds. Allow to stand 2 to 2-1/2 hours (no longer) with occasional shaking. Filter through a Whatman No. 42 filter paper. Transfer the precipitate from the flask (and the stopper) with 1% potassium nitrate solution, using a policeman. Wash the flask thoroughly and then wash the filter paper and precipitate, starting from the very top of the paper, until the washings are neutral to litmus (about 12 to 15 washings will be required). Place the paper and precipitate back into the original flask and add 40.00 ml of standard 0.3N sodium hydroxide solution, around the sides of the flask, with a pipet. Stopper the flask and swirl until the precipitate has dissolved. Wash down the stopper and sides of the flask and dilute to about 100 ml with CO_2-free water. Add 2 drops of phenolphthalein indicator solution (1%), and titrate with 0.15N nitric acid to the disappearance of the pink color. Ascertain the ml of sodium hydroxide equivalent to 1 ml of nitric acid by pipetting 40.00 ml of the standard 0.3N sodium hydroxide into a flask and titrating with nitric acid.

Calculate the percent phosphorus as follows:

$$\% \text{ phosphorus} = \frac{(A-BC)N \times 0.001360}{S} \times 100$$

where: A = ml of sodium hydroxide solution added to the sample.
B = ml of nitric acid required to titrate sample.
C = ml of sodium hydroxide equivalent to 1 ml of nitric acid.
N = normality of sodium hydroxide solution.
S = gm. of sample in aliquot.

4.5.5 Monoisopropyl dihydrogen phosphite.

4.5.5.1 Reagents.

4.5.5.1.1 Titration solvent. Add 500 ml of toluene and 5 ml of water to 495 ml of anhydrous isopropyl alcohol.

4.5.5.1.2 Standard alcoholic potassium hydroxide solution (0.1N). Add 6 gm. of potassium hydroxide to approximately 1 liter of anhydrous isopropyl alcohol. Boil gently for 10 minutes to effect solution. Allow to stand for 2 days and then filter the supernatant liquid through a fine sintered-glass funnel. Store the solution in a Pyrex bottle. Standardize against potassium acid phthalate.

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4.5.5.2 Procedure. Transfer 2 to 3 gm. of sample, accurately weighed, to a 250-ml beaker, add 100 ml of titration solvent, and stir to dissolve. Add 5 drops of methyl purple indicator and titrate to a green color with standard alcoholic potassium hydroxide solution (0.1N).

Calculate the percent monoisopropyl dihydrogen phosphite as follows:

$$\% \text{ monoisopropyl dihydrogen phosphite} = \frac{0.1242VN}{W} \times 100$$

where: V = ml of potassium hydroxide solution.
N = normality of potassium hydroxide solution.
W = gm. of sample.

4.5.6 pH. Dilute 1 ml of the sample to 100 ml with water that has been freshly boiled and has a pH of 6.0 ± 0.5. Measure the pH with a pH meter that is capable of measuring pH within 0.1 pH unit.

4.5.7 Chloride. Transfer 5 gm. of the sample to a 250-ml beaker and add 20 ml of 20% sodium hydroxide solution. Cover with a watch glass and allow to stand overnight. Dilute to about 150 ml and boil gently for 20 minutes. Allow to cool somewhat, neutralize with nitric acid using litmus paper, and add about 2 ml excess nitric acid. Remove the litmus paper. If the solution is not clear filter and wash with water. Dilute to about 150 ml, heat to boiling, add 5 ml of silver nitrate solution (1%), and boil for about a minute. Allow to stand overnight in a dark place. Filter through a tared fine sintered glass crucible, and transfer and wash with 1% nitric acid solution. Dry at 150°C for 1 hour, cool in a desiccator and weigh.

Calculate the percent chloride as follows:

$$\% \text{ chloride} = \frac{0.247A}{W} \times 100$$

where: A = gm. of precipitate.
W = gm. of sample.

4.5.8 Specific gravity. Determine the specific gravity at 25°/4°C by use of a pycnometer or any other method accurate to 3 decimal places.

4.5.9 Refractive index. Determine the refractive index at 20°C by use of a refractometer.

5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging will be levels A, B, or C as specified (see 6.2).

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5.1.1 Levels A and B. Diisopropyl hydrogen phosphite shall be packaged in accordance with MIL-STD-290 in unit quantities as specified in contract or order (see 6.2).

5.1.2 Level C. The diisopropyl hydrogen phosphite shall be packaged to afford the minimum degree of protection necessary to prevent damage during shipment under environmental conditions and commercial modes of transportation.

5.2 Packing. Packing shall be Level A, B. or C as specified (see 6.2).

5.2.1 Levels A and B. Diisopropyl hydrogen phosphite shall be packed in accordance with MIL-STD-290 (see 6.2).

5.2.2 Level C. Packages that require overpacking for acceptance by the common carrier shall be packed in exterior type shipping containers in a manner that will assure safe transportation at the lowest cost to the point of delivery and shall meet, as a minimum, the requirements of the rules and regulations to the mode of transportation selected.

5.3 Marking. In addition to any marking specified in 5.3 and special marking required by the contract or order, unit and intermediate containers and exterior shipping containers shall be marked in accordance with MIL-STD-129.

5.3.1 Precautionary marking. Each can or bottle and shipping container shall be durably and legibly labelled to show the following precautionary information:

WARNING!

Do not breathe vapors.
Use with adequate ventilation.
Avoid contact wiht skin and eyes. If liquid should contact skin or eyes, flush immediately with water for 15 minutes; for eyes, get medical attention.

6. NOTES

6.1 Intended use. Diisopropyl hydrogen phosphite is intended for use as an antiwear additive in lubricants.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification.
b. Quantity.
c. Selection of applicable level of packaging and packing required (see 5.1 and 5.2).
d. Special marking other than specified (see 5.3).

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6.3 Significant places. For the purpose of determining conformance with this specification, an observed or calculated value should be rounded off "to the nearest unit" in the last right-hand place of figures used in expressing the limiting value, in accordance with the rounding-off method of the Recommended Practices for Designating Significant Places in Specified Limiting Values (ASTM E 29).

Custodians:
Army - MU

Preparing activity:
Army - MU

Review activities:
Army - MU, ME, MD, MR, WC

Project No. 6810-AA19

User activities:
Army - AT, AV

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
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DOCUMENT IDENTIFIER AND TITLE
MIL-D-48499(MU) Diisopropyl Hydrogen Phosphite, Technical

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A
☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?
   ☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

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DATE

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